Rungus Morphological Analyzer

Technical Progress Report for Christine Dreiheller (SIL)

Overview

This report documents the current state of a prototype morphological analyzer for the Rungus language (ISO 639-3: drg), built as a response to the idea of developing a 'hermit crab parser' for Rungus. The analyzer decomposes inflected surface words into their base roots, prefixes, infixes, suffixes, and enclitics, applying morphophonological rewrite rules derived from Forschner's grammar and the Swarthmore LING073 transducer project.

Data sources:

- Webonary Rungus dictionary: 12,530 entries (scraped and normalized)
- Forschner, T.A. (1994). Outline of a Momogun Grammar (Rungus Dialect)
- Swarthmore College LING073 Spring 2025 Rungus transducer documentation
- Corpus: 534,775 tokens across 9 Rungus books (ebfo.de)

Current token coverage: 96.4% (515,762 of 534,775 tokens).
Live prototype: https://rungus-analyzer.vercel.app/

1. False Positives - Wrong Root Selected

The most common failure mode is a spelling collision: the analyzer strips the affixes correctly, but lands on the wrong root because two valid roots produce the same surface form after nasal substitution. The analyzer has no semantic understanding, so it relies entirely on dictionary structure and morphophonological rules to resolve ambiguity.

1.1 Nasal Substitution Collisions

When the transitive prefix mo- (or ma-) attaches to a stem beginning with a voiceless consonant (/p/, /t/, /k/, /s/) or /b/, the initial consonant is replaced by a nasal at the same place of articulation (p/b -> m, t/s -> n, k -> ng). During analysis, the engine must reverse this: an initial m- could come from p-, b-, v-, or an original m-.

Example: mamasa

```
Root: basa  -  "read"
Prefix: ma  (actor focus (a-vowel voiceless C stem) (transitive))
```

The engine strips ma- and recovers masa. Under nasal substitution rules, masa could come from basa ("to read"), pasa ("rotten"), or vas. The dictionary lists mamasa as a subentry of basa, so the engine uses this parent-child relationship to prefer basa. This fix works when the dictionary hints at the correct parent, but does not help when no subentry entry exists for the surface form.

1.2 The ko-...-o Nominalizing Circumfix

Rungus has two homophonous suffixes written -o: the verbal imperative patient focus (-o) and the nominal abstract noun suffix that forms a circumfix with ko- (ko-...-o). Without considering the prefix context, the engine cannot distinguish them. I implemented a rule: when ko- is the prefix and -o is the suffix, treat it as the nominalizing circumfix.

Example (fixed): kosunduvo

```
Root: sundu  -  "supernatural power, spirit"
Prefix: ko  (abstract noun nominalizer (ko-...-o circumfix) (nominal))
Suffix: -o  (abstract noun nominalizer (ko-...-o, /v/ glide before vowel-final root))
```

Example (fixed): konoruvo

```
Root: naru  -  "long"
Prefix: ko  (abstract noun nominalizer (ko-...-o circumfix) (nominal))
Suffix: -o  (abstract noun nominalizer (ko-...-o, /v/ glide before vowel-final root))
```

In kosunduvo, the root sundu ends in the high vowel /u/. Rungus inserts a /v/ glide to bridge the adjacent vowels before appending -o: ko- + sundu + o -> kosunduvo. The same pattern appears in konoruvo (anaru "long" -> konoruvo "length").

Questions for confirmation:

- Is the ko-...-o circumfix fully productive in Rungus?
- Does the /v/ glide insertion also occur after /i/, or only after /u/?
- Are there other circumfixes with similar morphophonological behavior?

1.3 False Reduplication Stripping

Words like kakal ("still/yet"), kakau ("epilepsy"), and kakat ("pull/drag") were initially misparsed: the engine saw the repeated ka- syllable and interpreted it as CV-reduplication, stripping it to produce false bases (kal, kau, kat). I added a guard that checks the full word in the dictionary before attempting any reduplication stripping.

2. False Negatives - Root Missing From Dictionary

The second failure mode is straightforward: a word fails to parse because its root is simply absent from the Webonary dictionary. No amount of morphological analysis can recover a root that was never documented.

Example: pogibad (85 occurrences, meaning unknown)

```
Root: (not found)
```

I have been working on a Human-in-the-Loop (HITL) workflow to address this: an AI examines the surrounding sentence context and pre-fills a spreadsheet with educated guesses about the root and meaning, which a human reviewer can then confirm or correct. The top 500 unanalyzed words are included as a separate CSV attachment. The web app also has an "Export Unanalyzed to CSV" button in Batch Mode for generating these lists from any Rungus text.

3. Morphophonological Rules Implemented

3.1 Nasal Substitution & Vowel Assimilation

The analyzer reverses consonant substitution for all voiceless stops and /b/: p->m, t/s->n, k->ng. It also handles prenasalization before voiced consonants (liquids, glides) where a nasal is inserted rather than substituted.

Example: mamanau (panau + ma-)

```
Root: panau  -  "walk"
Prefix: ma  (actor focus (a-vowel voiceless C stem) (transitive))
```

Example: monguvo (kuvo + mo-)

```
Root: kuvo  -  "What's his name"
Prefix: mo  (actor focus (voiceless C / b stem) (transitive))
```

Example: mangaradu (radu + ma- + prenasalization)

```
Root: radu  -  "plough"
Prefix: manga  (actor focus (a-stem voiced C) (transitive))
```

3.2 Vowel Contractions

When a vowel-ending prefix attaches to a vowel-initial stem, the two vowels contract: a+i -> e, o+i -> e, o+u -> u. The analyzer reverses these contractions during analysis.

Example: elo (ilo + o-, o+i -> e)

```
Root: ilo  -  "know, clever"
```

3.3 Glottal Stop Alternations

Root-final glottal stops alternate when a suffix is appended. After the high vowel /i/, the glottal becomes /z/. After /u/, it becomes /h/. After the low vowel /a/, it is retained. The analyzer reverses all three alternations.

Example: gilizon (gili' + -on, /i/ -> /z/)

```
Root: gili  -  "put"
Suffix: -on  (patient focus / gerundive (verbal))
```

Example: ontoluhan (ontolu' + -an, /u/ -> /h/)

```
Root: ontolu  -  "egg"
Suffix: -an  (reference focus / locative (verbal))
```

3.4 Non-Syllabic Vowel Consonantization

Word-final non-syllabic vowels become consonants when suffixes are appended: /u/ -> /h/ and /i/ -> /z/. The analyzer reverses these by trying h->u and z->i (or z-stripping for glottal stop origins) during root lookup.

Example: pasalahon (asalau + pa- + -on, /u/ -> /h/)

```
Root: asalau  -  "thick, strong (of rope, root of plant)"
Prefix: pa  (causative action (a-stem) (causative))
Suffix: -on  (patient focus / gerundive)
```

3.5 Infixation

Rungus uses three infixes: -um- (active voice), -in- (past tense), and -inum- (past intransitive, a contraction of -in- + -um-). Infixes are inserted after the first consonant of the root, or after the first consonant of the prefix when a prefix is present.

Example: riumikot (rikot + -um-)

```
Root: rikot  -  "come"
```

Example: rinumikot (rikot + -inum-)

```
Root: rikot  -  "come"
Infix: -inum-  (past of intransitive process)
```

3.6 Stacked Prefixes & Extreme Agglutination

The analyzer handles two layers of stacked prefixes and also tries suffixes on the final remainder. This allows it to parse words combining deep prefix stacking with suffixes.

Example: nangasavatan (na- + anga- + savat + -an)

```
Root: savat  -  "height"
Prefix: na  (patient focus perfect (a-stem) (perfect))
Stacked prefix: anga  (plural marker (a-stem) (plural))
Suffix: -an  (reference focus / locative (verbal))
```

3.7 Enclitics & Complete Predicate Words

Rungus attaches pronominal enclitics to the end of verbs, forming complete predicate sentences within a single word. The analyzer strips these before morphological decomposition and correctly identifies the subject pronoun.

Example: minonudukoku (tuduk + mo- + -in- + -oku "I")

```
Root: tuduk1  -  "teach"
Infix: -in-  (past / perfective marker)
Enclitic: -oku  (I / me (1sg subject) (pronominal))
```

Example: tinumorimaoku (torima + -inum- + -oku "I")

```
Root: torima  -  "receive"
Infix: -inum-  (past of intransitive process)
Enclitic: -oku  (I / me (1sg subject) (pronominal))
```

4. Open Questions for Expert Review

1. Is the ko-...-o nominalizing circumfix fully productive? Are there restrictions on which roots it can attach to?

2. Does the /v/ glide insertion before -o occur only after /u/, or also after /i/? The data shows it after /u/ (sundu -> kosunduvo, anaru -> konoruvo) but I have not found a clear /i/ example.

3. For the nasal substitution collision problem (e.g., mamasa -> basa vs. pasa), is there a frequency or semantic preference rule in Rungus that could help disambiguate?

4. Are there morphophonological rules I have not implemented that would be important for accurate analysis? In particular, I would like to verify my treatment of glottal stop alternations and non-syllabic vowel consonantization against your knowledge of the grammar.

5. For the missing-root words (Section 2), would you be able to review even a small portion of the attached spreadsheet? Even 20-30 entries would significantly improve coverage.

I would greatly appreciate the opportunity to meet in person to discuss these questions. Please feel free to reach me by email at aifvennelson08@gmail.com.

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